QUESTION ANALYSIS

EEE329

Communication Engineering I

Question Analysis

Sessions 2021–22 to 2020–21 *2 papers, 51 questions, 90 formulas*

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1. At a glance:

This analysis covers **2 papers** (2021-22, 2020-21), with 51 questions and 293 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Amplitude modulation	57	19 %	10	AM modulation index (19); DSB-SC (14); VSB (13)
Sampling and pulse modulation	48	16 %	9	PCM (20); Delta modulation (16); Sampling (12)
Multiplexing and multiple access	46	16 %	9	TDM (19); Multiple access (16); Spread spectrum (6)
Digital modulation	43	15 %	7	QAM (14); ASK, FSK and PSK (11); DPSK (10)
Angle modulation	29	10 %	5	Deviation and index (15); FM and PM (5); NBFM (5)

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2021-22	2020-21	All
Signals and spectrum	**13**	**6**	**19**
Propagation modes	**7**	·	7
EM spectrum	·	**6**	6
Distortionless transmission	**6**	·	6
Noise and channel capacity	**6**	**20**	**26**
Channel capacity	**6**	**16**	22
Thermal and transit-time noise	·	**4**	4
Amplitude modulation	**40**	**17**	**57**
AM modulation index	**15**	**4**	19
DSB-SC	**14**	·	14
VSB	**6**	**7**	13
Sidebands and power	·	**6**	6
Envelope detector	**5**	·	5
Angle modulation	**10**	**19**	**29**
Deviation and index	**5**	**10**	15
FM and PM	·	**5**	5
NBFM	**5**	·	5
Carson's rule	·	**4**	4
Receivers	·	**7**	**7**
Superheterodyne	·	**7**	7
Sampling and pulse modulation	**26**	**22**	**48**
PCM	**15**	**5**	20
Delta modulation	**6**	**10**	16
Sampling	**5**	**7**	12
Digital modulation	**24**	**19**	**43**

Topic and subtopic	2021-22	2020-21	All
QAM	**7**	**7**	14
ASK, FSK and PSK	**11**	·	11
DPSK	**6**	**4**	10
FSK	·	**8**	8
Multiplexing and multiple access	**18**	**28**	**46**
TDM	**12**	**7**	19
Multiple access	**6**	**10**	16
Spread spectrum	·	**6**	6
FDM	·	**5**	5
Information theory	**6**	**5**	**11**
Entropy	**6**	**5**	11
System design	**7**	·	**7**
Link design	**7**	·	7

3. Priority list:

Must know: Amplitude modulation (57), Sampling and pulse modulation (48), Multiplexing and multiple access (46), Digital modulation (43), Angle modulation (29), Noise and channel capacity (26), Signals and spectrum (19), Information theory (11)

Likely: Receivers (7), System design (7)

Less often: none

The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

4. Topic pages:

4.1. Signals and spectrum:

19 marks, 3 questions, in 2 of 2 papers. Book: Lathi Ch. 1, 3; Forouzan Ch. 3.

Most often theory (2 of 3). Subtopics: EM spectrum; Propagation modes; Distortionless transmission.

Formulas:

Subtopic	Formula	Equation	Used in
Propagation modes (7)	Radio horizon (LOS)	$d \approx 4.12\left(\sqrt{h_t}+\sqrt{h_r}\right)$ km	21-22 1(b)
Distortionless transmission (6)	Distortionless output	$y(t) = k\,x(t-t_d)$	21-22 1(d)
	Distortionless channel	$H(f) = k\,e^{-j2\pi f t_d}$	21-22 1(d)
	Group delay	$t_g = -\frac{1}{2\pi}\frac{d\theta_h}{df} = t_d$	21-22 1(d)

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Symbols:

h_t, h_r = antenna heights (m), $\frac{4}{3}$-Earth radius ↩

k = constant gain ↩

t_d = constant delay (s) ↩

θ_h = phase response (rad) ↩

Questions:

No.	Paper	Marks	Type	Question
EM spectrum				
1	20-21 1(b)	6	theory	Briefly explain the three major bands of the electromagnetic (EM) spectrum-radio waves, microwaves and infra-red waves-in terms of their bandwidth, propagation mode and applications.
Propagation modes				
2	21-22 1(b)	7	theory	Discuss the different modes of propagation of EM wave.
Distortionless transmission				
3	21-22 1(d)	6	derivation	Explain the conditions required for distortionless transmission.

4.2. Noise and channel capacity:

26 marks, 4 questions, in 2 of 2 papers. Book: Lathi Ch. 1, 13; Forouzan Ch. 3.

Most often numerical (3 of 4). Subtopics: Thermal and transit-time noise; Channel capacity.

Formulas:

Subtopic	Formula	Equation	Used in
Thermal and transit-time noise (4)	**Thermal noise power**	$P_n = kTB$	20-21 1(c)
	Thermal noise voltage	$V_n = \sqrt{4kTBR}$	20-21 1(c)
Channel capacity (22)	Equally likely levels	$H = \log_2 L$	20-21 3(d)
	Information rate	$R = f_s H$ bit/s	20-21 3(d)
	Shannon capacity	$C = B \log_2 \left(1 + \frac{S}{N}\right)$	20-21 3(d)
	Decibels to ratio	$\frac{S}{N} = 10^{\text{dB}/10}$	20-21 5(b)
	Minimum SNR	$\text{SNR}_{min} = 2^{R/B} - 1$	20-21 5(b)
	Bandwidth for a rate	$B = \frac{R}{\log_2(1 + S/N)}$	20-21 3(d)
	Attenuation	$10 \log_{10} \frac{P_t}{P_r} = 20 \log_{10} \frac{V_t}{V_r}$ dB	21-22 1(c)
	Signal-to-noise ratio	$\frac{S}{N} = \left(\frac{V_s}{V_n}\right)^2$	21-22 1(c)
	Shannon capacity	$C = B \log_2 \left(1 + \frac{S}{N}\right)$ bit/s	21-22 1(c)
	SNR for a rate	$\frac{S}{N} = 2^{R/B} - 1$	21-22 1(c)

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Symbols:

T = temperature (K)

R = resistance (Ω); wanted data rate (bit/s) ↩

L = number of levels ↩

f_s = symbols per second ↩

B = channel bandwidth (Hz) ↩

S/N = plain power ratio ↩

P_t, P_r = sent and received power (W) ↩

V_t, V_r = sent and received amplitude (V) ↩

V_s, V_n = signal and noise amplitude (V), same resistance ↩

C = largest error-free rate (bit/s) ↩

Questions:

No.	Paper	Marks	Type	Question
Thermal and transit-time noise				
1	20-21 1(c)	4	theory	Write short note on- (i) Thermal Noise (ii) Transit Time Noise
Channel capacity				
2	21-22 1(c)	6	numerical	We have a communication channel with a bandwidth of 5 MHz. A signal with an amplitude of 10 V is transmitted, and the receiver detects a signal with an amplitude of 2 V and noise with an amplitude of 5 mV. (i) Calculate the attenuation. (ii) Determine the maximum channel capacity. (iii) To achieve a data rate of 100 Mbit s^{-1}, find the required signal level.
3	20-21 3(d)	10	numerical	An analog signal having 4 kHz bandwidth is sampled at 1.25 times the Nyquist rate, and each sample is quantized into one of 256 equally likely level. Assuming the samples to be statistically independent: (i) What is the information rate of the source? (ii) Can the output of this source be transmitted without error over the AWGN channel with a bandwidth of 10 kHz and S/N ratio of 20 dB? (iii) Find S/N ratio required for error-free transmission of part ii. (iv) Find the bandwidth required for an AWGN channel for an error free transmission of output of this source if S/N ratio is 20 dB
4	20-21 5(b)	6	numerical	A voice grade telephone channel has a bandwidth of 3400 Hz. If the signal to noise ratio (SNR) on the channel is 30 dB, determine the capacity of the channel. If the above channel is to be used to transmit 4.8 kbps of data, determine the minimum SNR required on the channel.

4.3. Amplitude modulation:

57 marks, 10 questions, in 2 of 2 papers. Book: Lathi Ch. 4.

Most often theory (4 of 10). Subtopics: AM modulation index; Sidebands and power; VSB; DSB-SC; Envelope detector.

Formulas:

Subtopic	Formula	Equation	Used in
AM modulation index (19)	AM wave	$s(t) = [E_c + E_m \sin \omega_m t] \sin \omega_c t$	20-21 1(d)
	Modulation index	$\mu = \dfrac{E_m}{E_c}$	20-21 1(d)
	Side frequencies	$f_c \pm f_m$, amplitude $\dfrac{\mu E_c}{2}$	20-21 1(d)
	AM wave	$\varphi_{AM}(t) = [A + m(t)] \cos \omega_c t$	21-22 3(d)
	Modulation index	$\mu = \dfrac{m_p}{A}$	21-22 3(d)
	AM antenna current	$I_t = I_c \sqrt{1 + \dfrac{\mu^2}{2}}$	21-22 2(a)

Subtopic	Formula	Equation	Used in
Sidebands and power (6)	Carrier power	$P_c = \dfrac{E_c^2}{2R}$	20-21 1(e)
	Total power	$P_t = P_c\left(1 + \dfrac{\mu^2}{2}\right)$	21-22 2(a), 20-21 1(e)
VSB (13)	**VSB receiver filter**	$H_o(f) = \dfrac{1}{H_i(f+f_c) + H_i(f-f_c)}$	21-22 2(d), 20-21 4(b)
	VSB signal	$\Phi_{VSB}(f) = [M(f+f_c) + M(f-f_c)]\,H_i(f)$	21-22 2(d)
	VSB filter condition	$H_i(f+f_c) + H_i(f-f_c) = \text{constant},\ \ \lvert f\rvert \le B$	21-22 2(d)
DSB-SC (14)	DSB-SC spectrum	$S(f) = \dfrac{A_c}{2}[M(f-f_c) + M(f+f_c)]$	20-21 5(d)
	Coherent detector product	$V(f) = \dfrac{A_c}{2}M(f) + \dfrac{A_c}{4}[M(f-2f_c) + M(f+2f_c)]$	20-21 5(d)
	Lowest carrier	$f_c \ge W$	20-21 5(d)
	Coherent detection with errors	$e_o(t) = m(t)\cos(\Delta\omega\, t + \theta)$	21-22 2(b)
	Costas I and Q outputs	$\frac{1}{2}m(t)\cos\theta,\ \ \frac{1}{2}m(t)\sin\theta$	21-22 2(c)
	Costas control signal	$\frac{1}{8}m^2(t)\sin 2\theta$	21-22 2(c)
Envelope detector (5)	Envelope detector RC	$\dfrac{1}{\omega_c} \ll RC < \dfrac{1}{2\pi B}$	21-22 3(b)

Symbols:

E_c, E_m = peak carrier and message (V) ↩

A = carrier amplitude ↩

ω_c = carrier frequency (rad/s) ↩

I_t = rms current with modulation (A) ↩

I_c = rms current of the carrier alone (A) ↩

R = load (Ω) ↩

P_t = total power ↩

P_c = carrier power (W) ↩

H_i = transmit filter ↩

$H_o(f)$ = receiver LPF, for $|f| \le B$ ↩

$M(f)$ = message spectrum ↩

$H_i(f)$ = VSB shaping filter ↩

W = message bandwidth ↩

$\Delta\omega$ = local-carrier frequency error (rad/s) ↩

θ = phase error (rad) ↩

RC = load time constant (s) ↩

Questions:

No.	Paper	Marks	Type	Question
AM modulation index				
1	21-22 2(a)	5	numerical	The antenna current of an AM transmitter is 8 A when only the carrier is sent. But it increases to 8.93 A when the carrier is modulated by a single sine wave. Find the percentage modulation. Determine the antenna current when the percentage of modulation changes to 0.8
2	21-22 3(d)	10	diagram	For the AM signal with $m(t)$ in Fig. 1 given below, sketch $\Phi_A M(t)$ for $\mu = 0.5, \mu = 1$, and $\mu = 1.5$.
3	20-21 1(d)	4	numerical	A carrier wave of frequency 10 MHz and peak value of 10 V is amplitude modulated by a 5 kHz sine wave of amplitude 6 V. Determine the modulation index, side frequencies and amplitude of side frequencies.

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No.	Paper	Marks	Type	Question
Sidebands and power				
4	20-21 1(e)	6	numerical	An audio frequency signal $10\sin(2\pi 500t)$ is used to amplitude modulated a carrier of $50\sin(2\pi 105t)$ *[sic: should be* $2\pi 10^5 t$*]*. Calculate: (i) Modulation index (ii) Side-band frequencies (iii) Total power delivered to the load of 600 Ω
VSB				
5	21-22 2(d)	6	derivation	Find the condition of sideband shaping filter for VSB modulation.
6	20-21 4(b)	7	derivation	Explain how the message signal is reconstructed from VSB signal.
DSB-SC				
7	21-22 2(b)	6	theory	Explain how phase and frequency error affect the coherent detection of AM.
8	21-22 2(c)	8	theory	With appropriate block diagram, briefly explain the working principle of Costas DSB-SC receiver.
9	20-21 5(d)	0	theory	Consider a message signal m(t) with the spectrum shown in Fig. 1. The message bandwidth W = 1 kHz. This signal is applied to a product modulator, together with a carrier wave $A_c\cos(2\pi f_c t)$, producing the DSBSC modulated wave s(t). This modulated wave is next applied to a coherent detector. Assuming perfect synchronism between the carrier waves in the modulator and detector, determine the spectrum of the detector output when: (i) the carrier frequency $f_c = 1.25$ kHz and (ii) the carrier frequency $f_c = 0.75$ kHz.
Envelope detector				
10	21-22 3(b)	5	theory	Write a short note on the linear diode detection (envelope detector) method with proper diagrams and leveling.

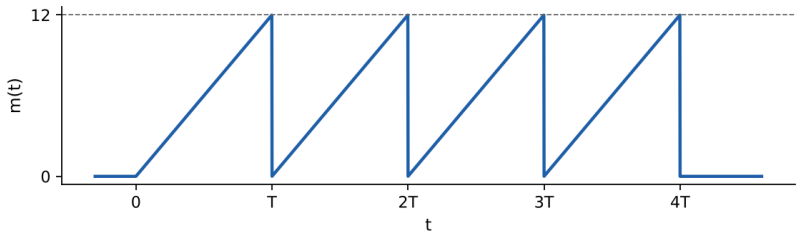


Figure 1: 21-22 3(d): Fig. 1 of the paper (redrawn): the message m(t)

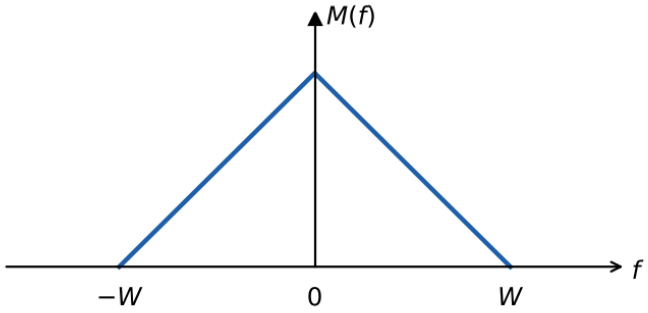


Figure 2: 20-21 5(d): Fig. 1 of the paper (redrawn)

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4.4. Angle modulation:

29 marks, 5 questions, in 2 of 2 papers. Book: Lathi Ch. 5.

Most often theory (3 of 5). Subtopics: FM and PM; Deviation and index; Carson's rule; NBFM.

Formulas:

Subtopic	Formula	Equation	Used in
FM and PM (5)	Instantaneous frequency	$\omega_i = \dfrac{d\theta}{dt}$	20-21 2(c)
	PM	$\varphi_{PM} = A\cos[\omega_c t + k_p m(t)]$	20-21 2(c)
	FM	$\varphi_{FM} = A\cos\left[\omega_c t + k_f \int m\, d\alpha\right]$	20-21 2(c)
Deviation and index (15)	**Frequency deviation**	$\Delta f = k_f V_m$	20-21 2(d)
	FM modulation index	$\beta = \dfrac{\Delta f}{f_m}$	21-22 3(c), 20-21 2(d)
	Frequency limits	$f_{max,min} = f_c \pm \Delta f$	21-22 3(c)
	Carrier swing	$\text{swing} = 2\Delta f$	21-22 3(c)
	Carson's rule	$B_T \approx 2(\Delta f + f_m)$	21-22 3(c)
Carson's rule (4)	**Carson's rule**	$B_T \approx 2(\Delta f + f_m) = 2(\beta + 1) f_m$	20-21 4(d)
NBFM (5)	FM wave	$\varphi_{FM}(t) = A\cos[\omega_c t + k_f a(t)],\ a(t) = \int m(\tau)\, d\tau$	21-22 3(a)
	NBFM	$\varphi_{NBFM}(t) \approx A\cos\omega_c t - A\, k_f\, a(t)\sin\omega_c t$	21-22 3(a)

Symbols:

$\theta(t)$ = carrier angle

k_p = phase sensitivity ↩

k_f = frequency sensitivity (rad/s per volt) ↩

f_m = modulating frequency (Hz) ↩

Δf = peak frequency deviation (Hz) ↩

B_T = FM bandwidth (Hz) ↩

Questions:

No.	Paper	Marks	Type	Question
FM and PM				
1	20-21 2(c)	5	theory	With proper block diagram explain the relation between frequency modulation and phase modulation.
Deviation and index				
2	21-22 3(c)	5	numerical	A 93.2 MHz carrier is frequency modulated by a 5 kHz sine wave. The resultant FM signal has a frequency deviation of 40 kHz. (i) Find the carrier swing of the FM signal. (ii) What are the highest and lowest frequencies attained by the frequency modulated signal? (iii) Calculate the modulation index for the wave.
3	20-21 2(d)	10	numerical	In an FM system, when the audio frequency is 500 Hz and modulating voltage 2.5 V, the deviation produced is 5 kHz. If the modulating voltage is now increased to 7.5 V, calculate the new value of frequency deviation produced. If the AF voltage is raised to 10 V while the modulating frequency dropped to 250 Hz, what is the frequency deviation? Calculate the modulation index in each case.
Carson's rule				
4	20-21 4(d)	4	theory	State Carson's rule and its limitations.

No.	Paper	Marks	Type	Question
NBFM				
5	21-22 3(a)	5	theory	Explain NBFM generation process with necessary block diagrams and equations.

4.5. Receivers:

7 marks, 1 question, in 1 of 2 papers. Book: Lathi Ch. 4.

Most often theory (1 of 1). Subtopics: Superheterodyne.

Formulas:

Subtopic	Formula	Equation	Used in
Superheterodyne (7)	**Local oscillator**	$f_{LO} = f_s + f_{IF}$	20-21 4(a)
	Image frequency	$f_{si} = f_s + 2f_{IF}$	20-21 4(a)

Questions:

No.	Paper	Marks	Type	Question
Superheterodyne				
1	20-21 4(a)	7	theory	With appropriate block diagram, describe the operation of superheterodyne receiver.

4.6. Sampling and pulse modulation:

48 marks, 9 questions, in 2 of 2 papers. Book: Lathi Ch. 6.

Most often theory (5 of 9). Subtopics: Delta modulation; PCM; Sampling.

Formulas:

Subtopic	Formula	Equation	Used in
Delta modulation (16)	**DM largest slope**	σf_s	20-21 3(a)
	No slope overload	$\left\lvert \frac{dm}{dt} \right\rvert_{max} \leq \sigma f_s$	20-21 3(a)
PCM (20)	Bits per sample	$n = \log_2 L$	20-21 3(c)
	PCM bit rate	$R_b = nf_s$	21-22 4(e), 20-21 3(c)
	Information rate	$R = r\,H$ bits/s	21-22 4(c)
	Step size	$\Delta = \frac{2m_p}{L}$	21-22 4(e)
	Bits per sample	$L = 2^n,\ n = \log_2 L$	21-22 4(e)
	Quantization noise	$N_q = \frac{\Delta^2}{12} = \frac{m_p^2}{3L^2}$	21-22 4(a)
	SQNR	$\text{SQNR} = 3L^2 \frac{\overline{m^2}}{m_p^2}$	21-22 4(a)
	Peak signal to peak noise	$\frac{S_{peak}}{N_{peak}} = L^2$	21-22 4(e)

Subtopic	Formula	Equation	Used in
	Minimum bandwidth	$B_T = \frac{R_b}{2} = nB$	21-22 4(e)
	μ-law compressor	$y = \frac{\ln(1+\mu\lvert m\rvert/m_p)}{\ln(1+\mu)}\operatorname{sgn}(m)$	21-22 4(c)
	Companded SQNR	$\text{SQNR} \approx \frac{3L^2}{[\ln(1+\mu)]^2}$	21-22 4(c)
Sampling (12)	**Nyquist rate**	$f_s \geq 2B$	21-22 4(b), 20-21 4(c)
	Sampled spectrum	$\bar{G}(f) = \frac{1}{T_s}\sum_n G(f - nf_s)$	20-21 4(c)
	Interpolation formula	$g(t) = \sum_k g(kT_s)\,\text{sinc}(2\pi Bt - k\pi)$	21-22 4(b), 20-21 4(c)
	Sampled spectrum	$\bar{G}(f) = \frac{1}{T_s}\sum_{n=-\infty}^{\infty} G(f - nf_s)$	21-22 4(b)
	Reconstruction LPF	$H(f) = T_s\,\text{rect}\left(\frac{f}{2f_0}\right),\ B \leq f_0 \leq f_s - B$	21-22 4(b)

Symbols:

σ = step size ↩

R_b = bit rate (bit/s) ↩

r = symbol rate (symbols/s) ↩

L = number of levels ↩

n = bits per code word ↩

$\overline{m^2}$ = message power ↩

μ = compression constant (255), not the AM index ↩

B = highest message frequency (Hz) ↩

f_s = sampling rate (samples/s) ↩

$G(f)$ = message spectrum

$T_s = 1/f_s$ = sampling interval (s)

f_0 = filter cutoff (Hz) ↩

Questions:

No.	Paper	Marks	Type	Question
Delta modulation				
1	21-22 5(a)	6	theory	Explain the principle of sigma-delta modulator. Also explain how it is advantageous over DM.
2	20-21 3(a)	5	theory	Define slope overload distortion and granular noise in delta modulation and how it is removed in ADM?
3	20-21 3(b)	5	diagram	Draw the transmitter and receiver circuits block diagram of delta modulation.
PCM				
4	21-22 4(a)	5	theory	What is PCM? Explain PCM system.
5	21-22 4(c)	5	theory	Write short note on: Companding, Information rate, and Entropy of a source.
6	21-22 4(e)	5	numerical	An audio signal is sampled at 8 kHz rate using a 32-level quantizer. Find- (i) Peak signal power to peak quantization noise power. (ii) System bandwidth.
7	20-21 3(c)	5	diagram	Explain the basic block diagram of PCM system.
Sampling				
8	21-22 4(b)	5	theory	”A proper valued LPF filter can be used to reconstruct the sampled signal”-justify the statement with the proper mathematical explanations.
9	20-21 4(c)	7	derivation	State and prove Nyquist sampling theorem. Also find the interpolation formula for the reconstruction of analog signals from its uniform samples.

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4.7. Digital modulation:

43 marks, 7 questions, in 2 of 2 papers. Book: Lathi Ch. 7, 10; Forouzan Ch. 5.

Most often theory (5 of 7). Subtopics: FSK; QAM; DPSK; ASK, FSK and PSK.

Formulas:

Subtopic	Formula	Equation	Used in
FSK (8)	FSK frequencies	$f_{1,0} = f_c \pm \Delta f$	20-21 6(a)
	FSK bandwidth	$B_{FSK} \approx 2\Delta f + 2R_b$	20-21 6(a)
	FSK bandwidth (Forouzan)	$B = (1+d)S + 2\Delta f$	21-22 4(d), 20-21 6(a)
QAM (14)	**QAM signal**	$\varphi_{QAM} = m_1 \cos\omega_c t + m_2 \sin\omega_c t$	20-21 6(b)
	QAM with phase error	$m_1 \cos\theta - m_2 \sin\theta$	20-21 6(b)
	QAM signal	$\varphi_{QAM}(t) = m_1(t) \cos\omega_c t + m_2(t) \sin\omega_c t$	21-22 5(b)
	QAM with phase error	$m_1(t) \cos\theta - m_2(t) \sin\theta$	21-22 5(b)
DPSK (10)	**DPSK encoder**	$q_k = I_k \oplus q_{k-1}$	21-22 5(c), 20-21 6(d)
	DPSK receiver	$y_k \propto a_k a_{k-1}$	21-22 5(c), 20-21 6(d)
ASK, FSK and PSK (11)	ASK and PSK bandwidth	$B = (1+d)S$	21-22 4(d)
	Coherent BASK output	$dA\cos^2\omega_c t \rightarrow \frac{dA}{2}$	21-22 5(d)

Symbols:

R_b = bit rate ↩

S = baud rate; signal (baud) rate ↩

Δf = half the spacing of f_1 and f_2 (Hz) ↩

m_1, m_2 = the two messages ↩

I_k = data bit ↩

q_k = encoded bit ↩

a_k = polar symbol (± 1) ↩

y_k = LPF output ↩

d = filter factor, $0 \leq d \leq 1$; data (1 or 0) ↩

Questions:

No.	Paper	Marks	Type	Question
FSK				
1	20-21 6(a)	8	theory	Briefly describe FSK generation with the necessary block diagram. Include its spectrum and discuss the bandwidth requirements, along with the advantages and disadvantages of FSK.
QAM				
2	21-22 5(b)	7	theory	With proper block diagram explain the principle of coherent detection of QAM. Mention its advantages and drawbacks.
3	20-21 6(b)	7	theory	With proper block diagram explain the principle of coherent detection of QAM. Mention its advantages and drawbacks.
DPSK				
4	21-22 5(c)	6	numerical	Tabulate the values of signal at each stage of DPSK encoder and receiver for message signal '1010010110'.
5	20-21 6(d)	4	numerical	Tabulate the values of signal at each stage of DPSK encoder and receiver for message signal '1010011100'
ASK, FSK and PSK				
6	21-22 4(d)	5	theory	Name three mechanisms for the modulation of digital data into an analog signal. Show the differences among them.

No.	Paper	Marks	Type	Question
7	21-22 5(d)	6	theory	Briefly describe coherent and non-coherent detection of BASK and BFSK.

4.8. Multiplexing and multiple access:

46 marks, 9 questions, in 2 of 2 papers. Book: Lathi Ch. 11; Forouzan Ch. 6, 12.

Most often theory (6 of 9). Subtopics: Multiple access; FDM; TDM; Spread spectrum.

Formulas:

Subtopic	Formula	Equation	Used in
FDM (5)	**FDM link bandwidth**	$B_{link} = NB_{ch} + (N-1)B_g$	20-21 2(b)
TDM (19)	Quantization step	$\Delta = \frac{2m_p}{L}$	20-21 5(c)
	Largest error	$e_{max} = \frac{\Delta}{2} = \frac{m_p}{L}$	20-21 5(c)
	TDM bit rate	$R_b = Nf_sn$	21-22 6(b), 20-21 5(c)
	Minimum bandwidth	$B_T = \frac{R_b}{2}$	20-21 5(c)
	Levels from an error limit	$\frac{m_p}{L} \leq \epsilon\, m_p \;\Rightarrow\; L \geq \frac{1}{\epsilon}$	21-22 6(b)
	Synchronous TDM link	$R_{link} = \sum R_i$	21-22 6(a)
Spread spectrum (6)	FHSS channels	$M = 2^k$, $B_{SS} \approx MB$	20-21 6(c)

Symbols:

N = number of channels ↩ m_p = peak signal ↩ R_i = input rates ↩

B_g = guard band ↩ ϵ = allowed error as a fraction of m_p ↩ k = PN pattern bits

Questions:

No.	Paper	Marks	Type	Question
Multiple access				
1	21-22 6(c)	3	theory	Compare between FDMA and CDMA.
2	21-22 6(e)	3	theory	Differentiate between multiplexing and multiple access technologies.
3	20-21 2(a)	5	theory	Differentiate between multiplexing and multiple access technologies.
4	20-21 5(a)	5	theory	Write down the comparison between TDMA and CDMA.
FDM				
5	20-21 2(b)	5	numerical	Five channels, each with 100 kHz bandwidth, are to be frequency division multiplexing together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channel to prevent interference?
TDM				
6	21-22 6(a)	6	theory	Write short notes on: Multilevel multiplexing and Multiple slot allocation.

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No.	Paper	Marks	Type	Question
7	21-22 6(b)	6	numerical	It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patients are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitudes is 0.25 % of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data.
8	20-21 5(c)	7	numerical	It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patents *[sic: should be "patients"]* are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitude is 0.25% of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data.
Spread spectrum				
9	20-21 6(c)	6	theory	Explain the principle of frequency hopping spread spectrum technique.

4.9. Information theory:

11 marks, 2 questions, in 2 of 2 papers. Book: Lathi Ch. 13.

Most often numerical (2 of 2). Subtopics: Entropy.

Formulas:

Subtopic	Formula	Equation	Used in
Entropy (11)	Information in a symbol	$I = \log_2 \frac{1}{P}$ bits	20-21 1(a)
	Entropy	$H = \sum_i P_i \log_2 \frac{1}{P_i}$ bits/symbol	20-21 1(a)
	Binary entropy	$H(p) = -p \log_2 p - (1-p)\log_2(1-p)$	21-22 1(a), 20-21 1(a)
	Information in a symbol	$I(m_i) = \log_2 \frac{1}{P_i}$ bits	21-22 1(a)
	Entropy	$H(m) = \sum_{i=1}^{n} P_i \log_2 \frac{1}{P_i}$ bits/symbol	21-22 1(a)
	Bounds of entropy	$0 \leq H(m) \leq \log_2 n$	21-22 1(a)

Symbols:

P = probability of the symbol ↩

H = average information per symbol ↩

p = probability of one symbol; probability of one of two symbols ↩

P_i = probability of symbol m_i ↩

n = number of symbols ↩

Questions:

No.	Paper	Marks	Type	Question
Entropy				
1	21-22 1(a)	6	numerical	Find the entropy $H(m)$ of a memoryless source.
2	20-21 1(a)	5	numerical	Find the entropy of single bit binary memoryless source. Also draw the curve of entropy Vs. probability for that source.

4.10. System design:

7 marks, 1 question, in 1 of 2 papers. Book: Lathi Ch. 1.

Most often theory (1 of 1). Subtopics: Link design.

Questions:

No.	Paper	Marks	Type	Question
Link design				
1	21-22 6(d)	7	theory	Discuss the design procedures of a communication system.

5. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
Signals and spectrum				
1	21-22 1(d)	6	Distortionless transmission	Write the output as a delayed scaled input; Fourier transform it; read magnitude and phase conditions.
Amplitude modulation				
2	21-22 2(d)	6	VSB	Write the VSB spectrum; shift by the carrier; keep the baseband terms; require a constant bracket.
3	20-21 4(b)	7	VSB	Shape DSB-SC with a filter; multiply by the carrier; filter; require the sum of the shifted responses to be constant.
Sampling and pulse modulation				
4	20-21 4(c)	7	Sampling	Multiply by an impulse train; write its Fourier series; read off the repeated spectrum; filter with an ideal LPF.

6. Repeat tracker:

No repeats among these papers.

7. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical	Diagram
Signals and spectrum	2	1	·	·
Noise and channel capacity	1	·	3	·
Amplitude modulation	4	2	3	1
Angle modulation	3	·	2	·
Receivers	1	·	·	·
Sampling and pulse modulation	5	1	1	2
Digital modulation	5	·	2	·
Multiplexing and multiple access	6	·	3	·
Information theory	·	·	2	·

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Topic	Theory	Derivation	Numerical	Diagram
System design	1	·	·	·

8. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Amplitude modulation	57	19 %
2	Sampling and pulse modulation	48	36 %
3	Multiplexing and multiple access	46	52 %
4	Digital modulation	43	66 %
5	Angle modulation	29	76 %
6	Noise and channel capacity	26	85 %
7	Signals and spectrum	19	91 %
8	Information theory	11	95 %
9	Receivers	7	98 %
10	System design	7	100 %

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